



Machine Vision with Machine Learning #1

FRA 626 Machine Vision in Smart Factory

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Topics

- Machine Vision vs Machine Learning
- Machine Learning in Computer Vision
- Artificial Neural Networks
- Convolutional Neural Networks

Machine Vision vs Machine Learning

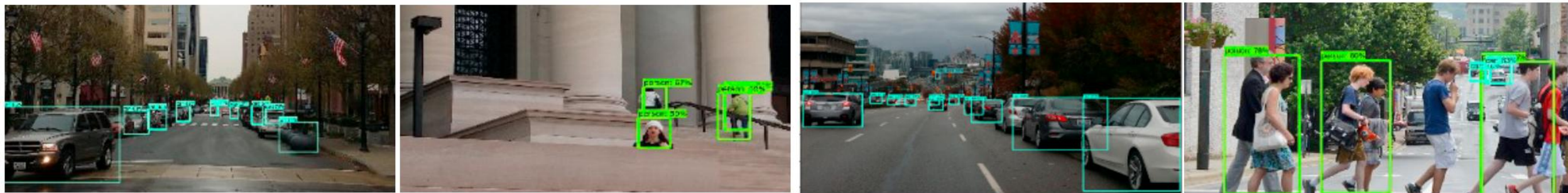
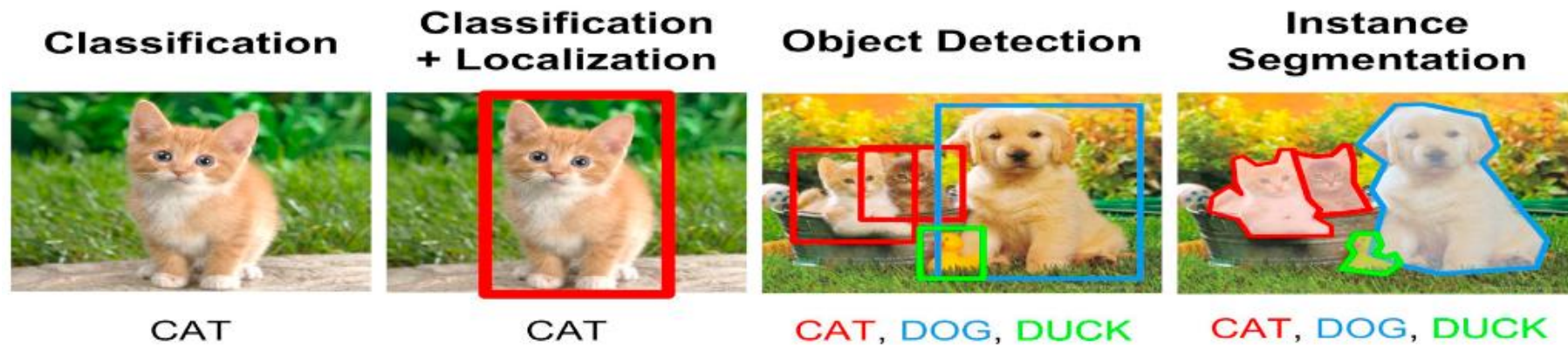
- Machine learning is programming technology to be able to adapt on its own.
- There are multiple techniques and strategies, but in the end the computer is able to use historical data while it functions.
- Machine learning goes along with artificial intelligence and is used often in modern manufacturing.
- Machine learning is the subset of artificial intelligence.

- Machine vision joins machine learning in a set of tools that gives consumer- and commercial-level hardware unprecedented abilities to observe and interpret
- Machine vision can be used with sensors, cobots, and other IoT technologies. their environment.

Autonomous vehicles Healthcare
Google Translate app
Facial recognition Real-time sports tracking
Agriculture Manufacturing

Machine Learning in Computer Vision

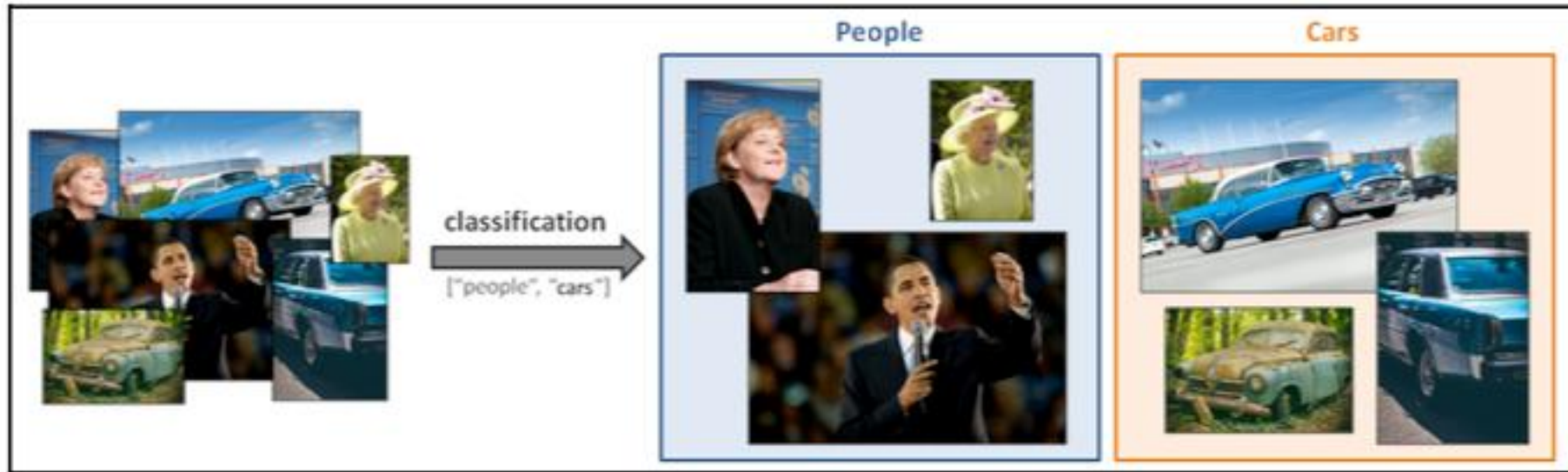
- Computer vision combined with machine learning for automatic analysis/annotation of videos.



Detecting cars and persons in images applying Deep learning and Faster-RCNN-Inception-V2 model

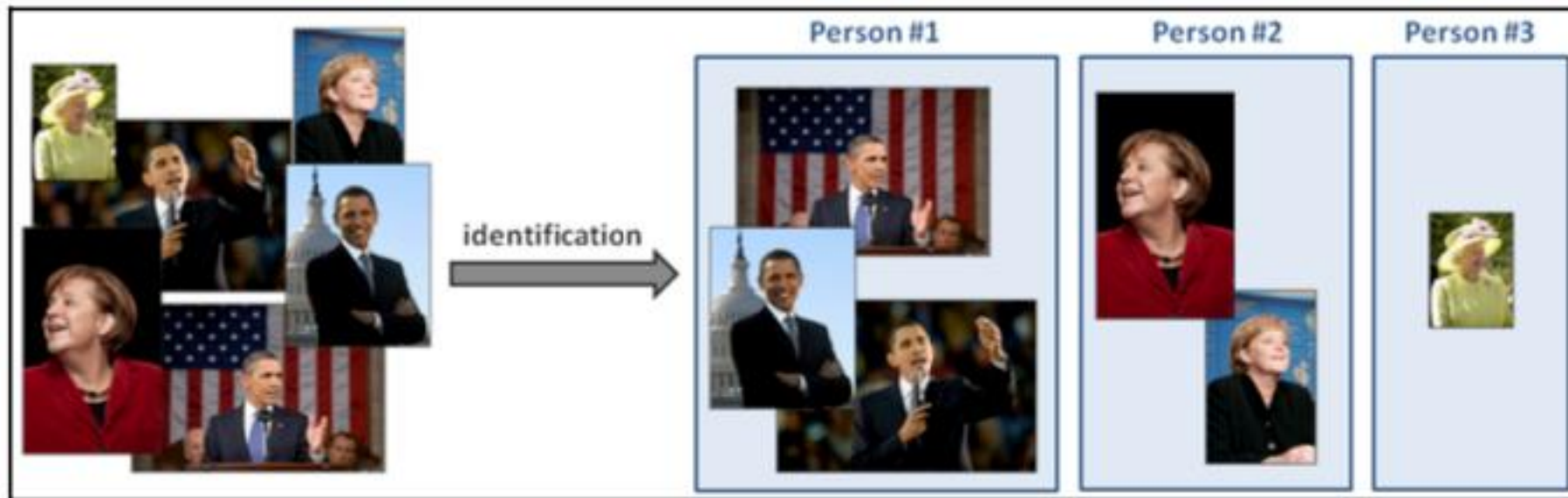
Object Classification

The task of assigning proper labels (or classes) to images among a predefined set



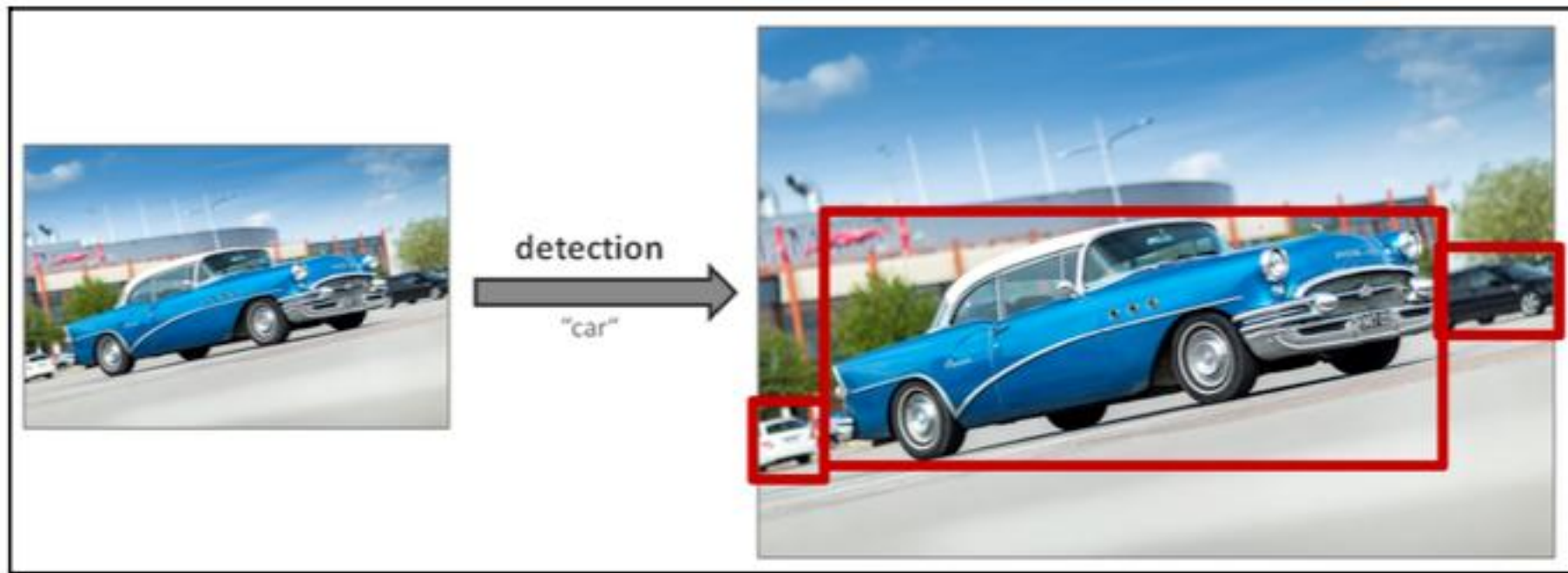
Object Identification

The methods assign labels from a predefined set, object identification (or instance classification) methods learn to recognize specific instances of a class.



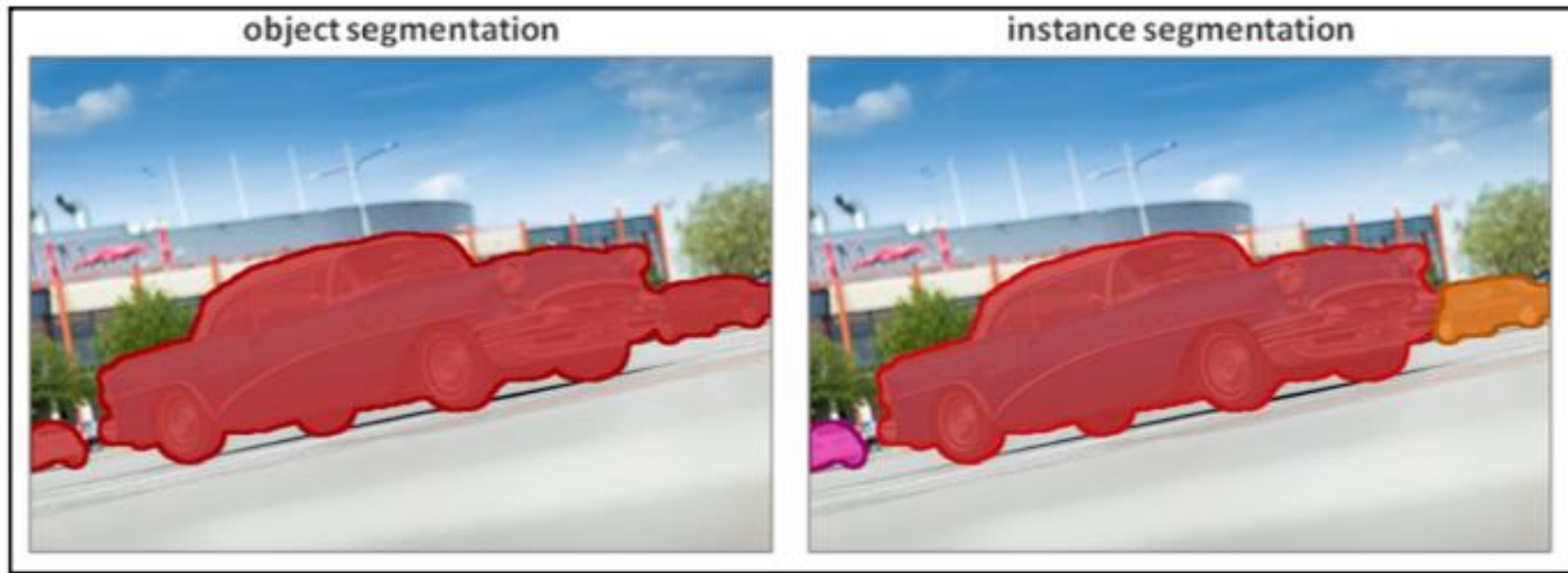
Object Detection and Localization

The task is the detection of specific elements in an image. It is often a preliminary step before further computations, providing smaller patches of the image to be analyzed separately.



Object and Instance Segmentation

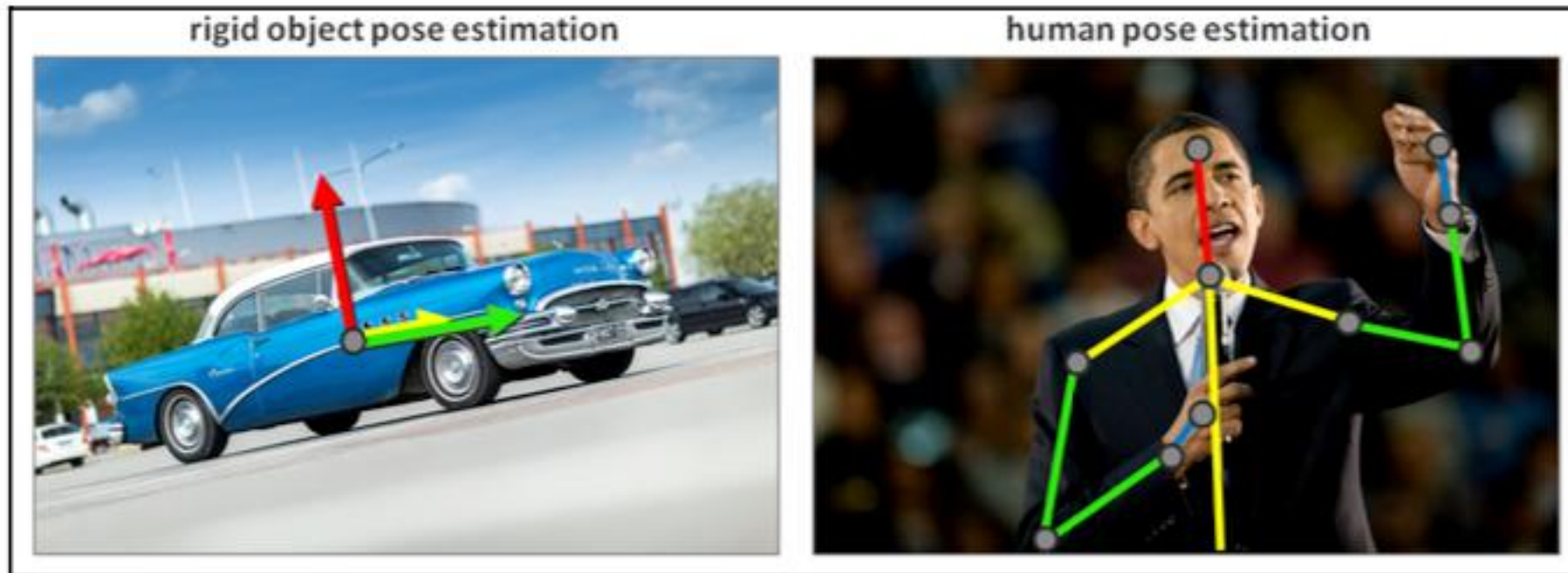
Segmentation can be seen as a more advanced type of detection. Instead of simply providing bounding boxes for the recognized elements, segmentation methods return masks labeling all the pixels belonging to a specific class or to a specific instance of a class.



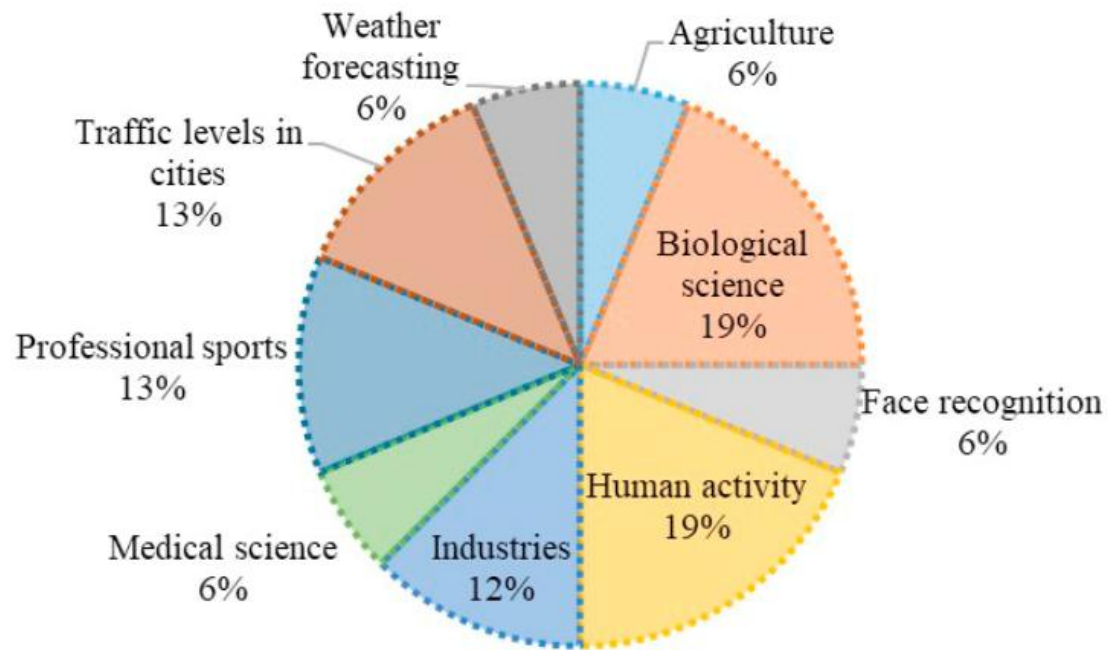
Pose Estimation

For rigid objects, pose estimation usually means the estimation of the objects' positions and orientations relative to the camera in the 3D space.

For non-rigid elements, pose estimation can also mean the estimation of the positions of their sub-parts relative to each other.



Machine Learning in Computer Vision



Machine learning and computer vision research areas

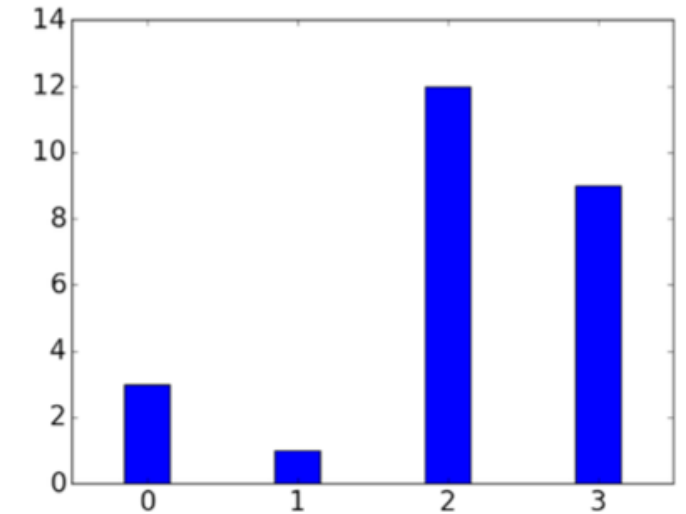


Frequently used themes of machine learning in computer vision

Feature Extraction

- Color Histogram
- Texture
 - GLCM (gray-level co-occurrence matrix)
 - GLGCM (gray-level gradient-based co-occurrence matrix)
 - LBP (local binary patterns)

3	2	2	0	3
1	3	0	2	2
2	2	2	2	3
3	3	3	2	3
0	2	3	2	2



	0	1	2	3
0	0	0	2	1
1	0	0	0	1
2	1	0	6	3
3	1	0	3	2

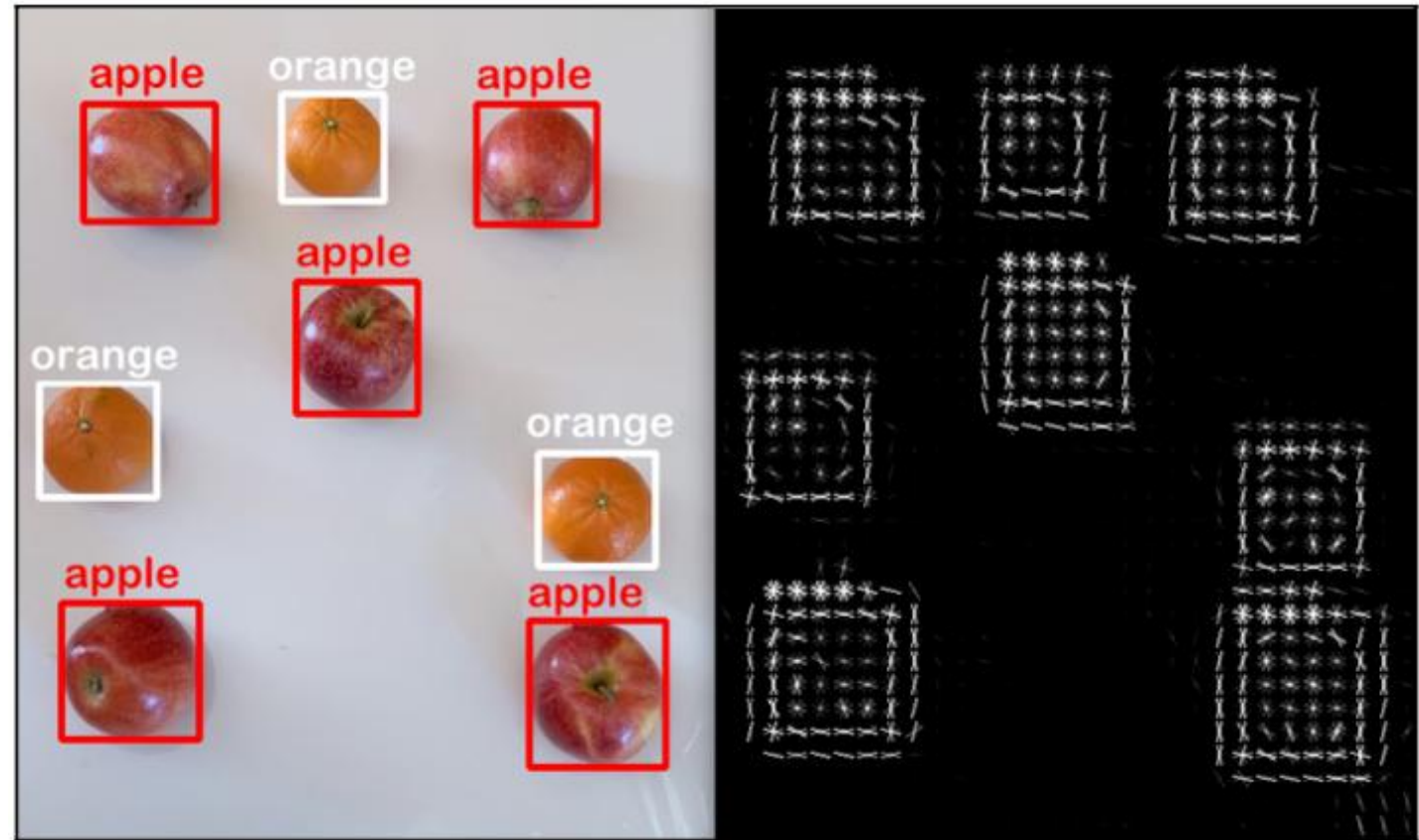
GLCM matrix with distance is 1 and angle 0

	0	1	2	3
0	0.0	0.0	0.1	0.05
1	0.0	0.0	0.0	0.05
2	0.05	0.0	0.3	0.15
3	0.05	0.0	0.15	0.1

Normalized GLCM

Feature Extraction

- Edge
 - HOG (Histogram of Oriented Gradients)
 - The HOG descriptor focuses on the structure or the shape of an object by extracting the gradient and orientation (or you can say magnitude and direction) of the edges.
 - The complete image is broken down into smaller regions and for each region, the gradients and orientation are calculated.
 - The histogram for each of these regions are created using the gradients and orientations of the pixel values



A. F. Gad, Practical Computer Vision Applications Using Deep Learning with CNNs

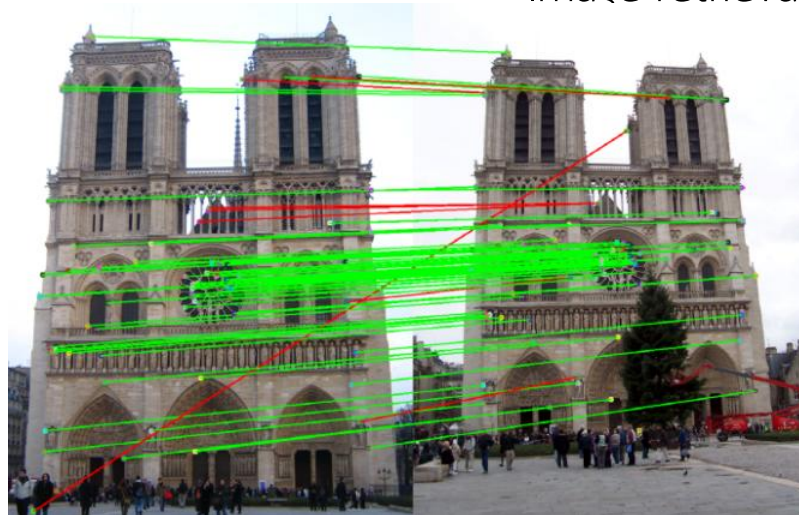
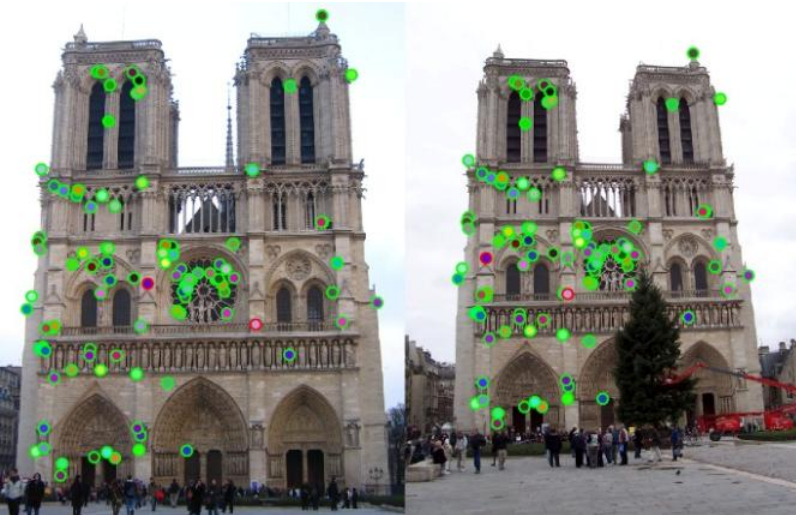


Other Feature Extraction

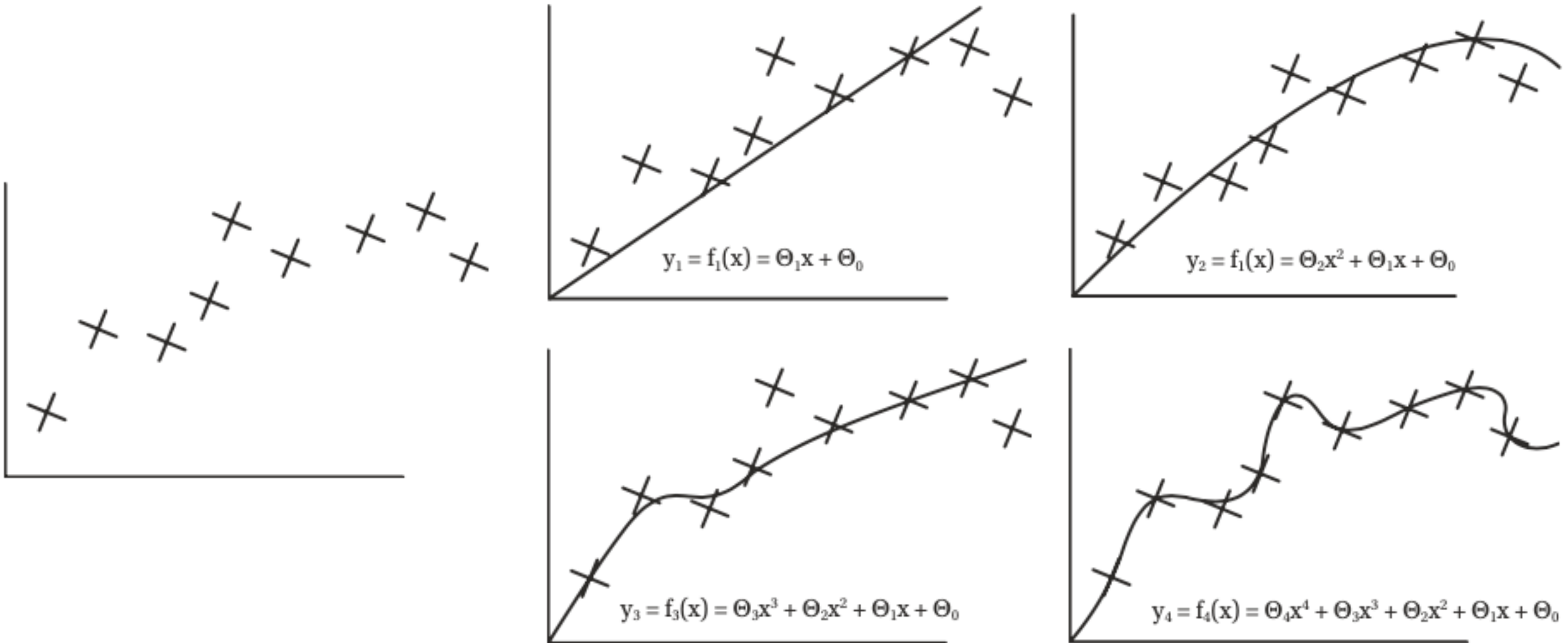
- Harris Corner
- SIFT (Scale Invariant Feature Transform)
- SURF (Speeded Up Robust Feature)
- FAST (Features from Accelerated Segment Test)
- ORB (Oriented FAST and Rotated BRIEF)

Application

- Automate object tracking
- Point matching for computing disparity
- Stereo calibration (Estimation of the fundamental matrix)
- Motion-based segmentation
- Recognition
- 3D object reconstruction
- Robot navigation
- Image retrieval and indexing



Regularization Based on a Regression Example

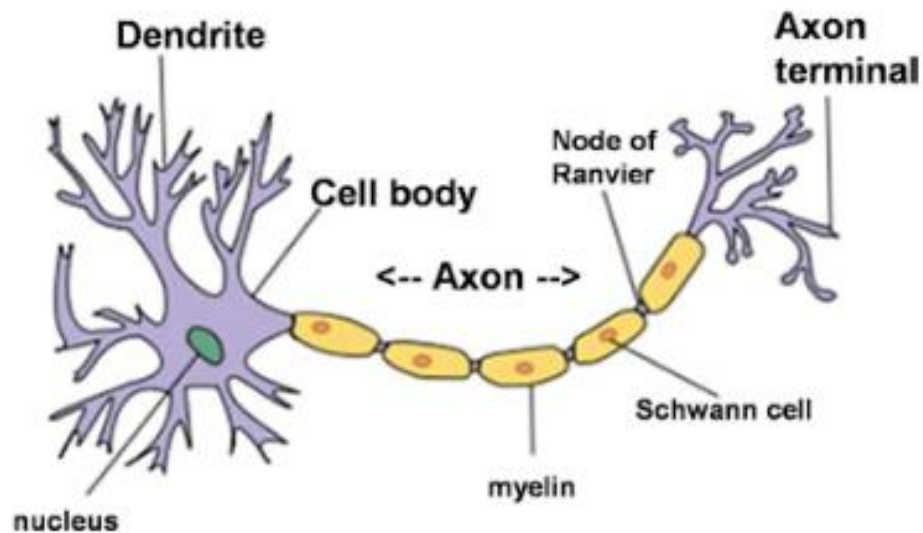


Artificial Neural Networks

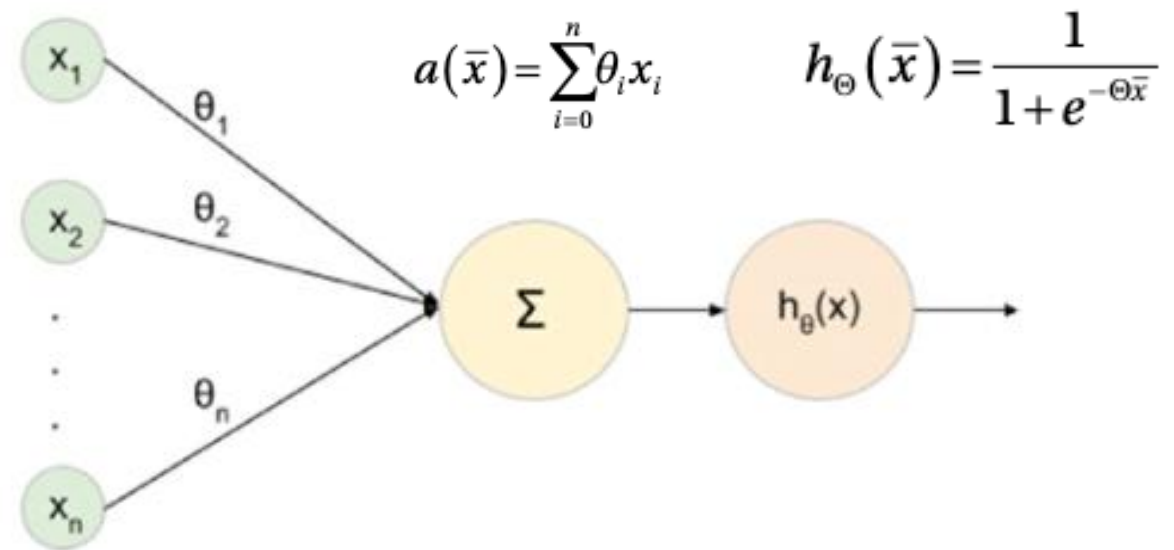
- Machine learning (ML) problems can be divided into three categories: supervised, unsupervised, and reinforcement.
- The supervised learning algorithm explores the data collected from experiments to map inputs to outputs.
- The unsupervised learning will not be given the results of the experiments nor a metric. There is no human to guide it at all. This is very challenging.
- In reinforcement learning, the human gives the robot a metric to evaluate its performance. The robot has to maximize this metric to reach its goal.
- An artificial neural network (ANN) is a kind of algorithm that works in all of these problems.
- ANN is a biologically inspired ML model that mimics the operation of the human brain.
- ANN is one of the most important topics to be covered when talking about deep learning (DL).

Artificial Neural Networks

- Neural networks come as a better approach to solve complex regression or classification problems with non-linear dependencies between the input features.



Principal components of a physical neuron



Logic model of a network with a single neuron (perceptron)

Artificial Neural Networks

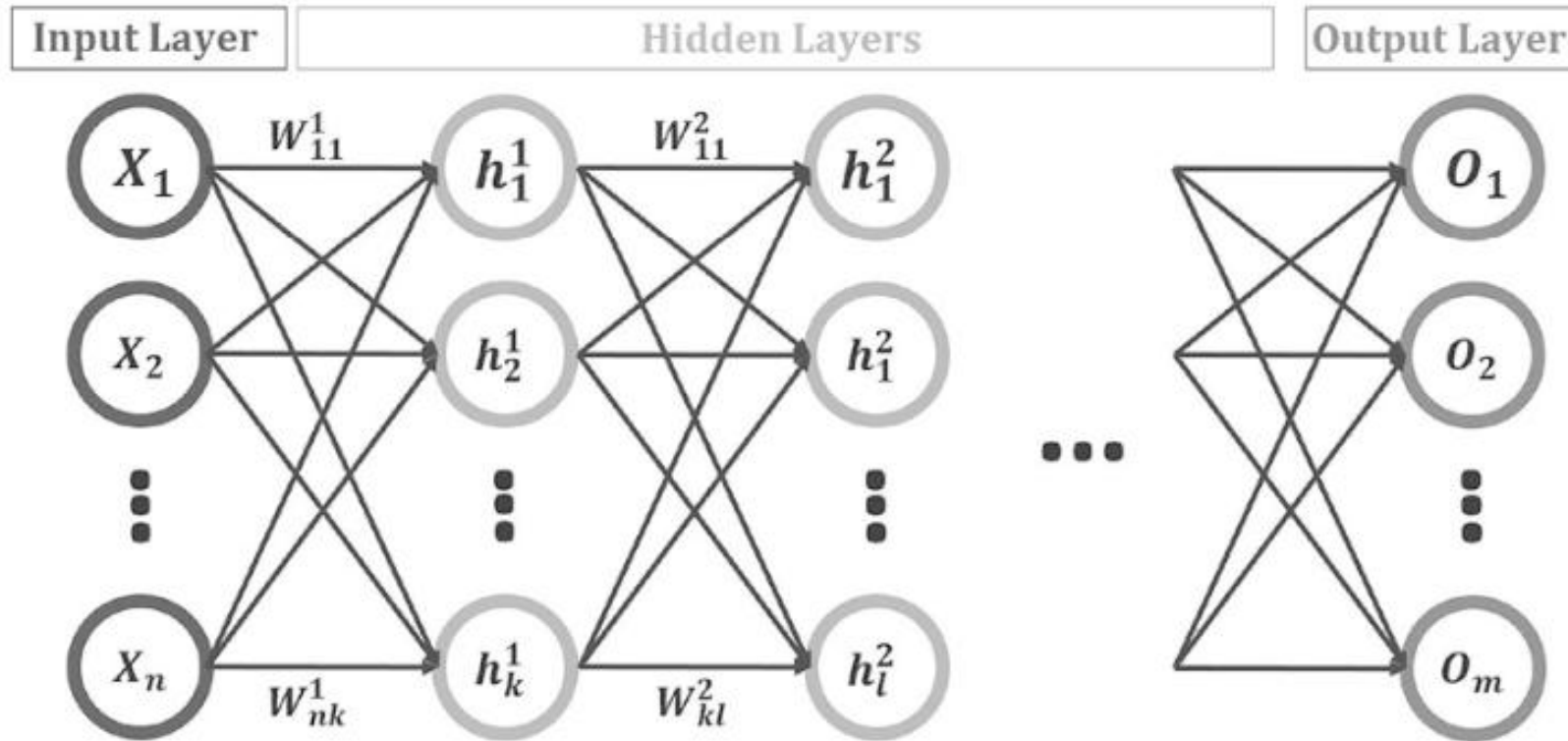


Figure 2-7. General FC artificial network architecture

Error Metrics

- **Mean squared error**, often used as the driving cost function of the model in regression problems
- **Mean absolute error**, sometimes used as an additional performance metric
- **Classification error**, used as a driving cost function both in logistic regression and neural networks
- **Accuracy**, defined as number of correct classifications divided by the total number of samples, and probably one of the most popular performance metrics

Confusion Metrics

- True positives (TP):
Data points labelled as positive and predicted as positive
($y = 1$ and $hQ(x) = 1$)
- True negatives (TN):
Data points labelled as negative and predicted as negative
($y = 0$ and $hQ(x) = 0$)
- False positives (FP):
Data points labelled as negative but predicted as positive
($y = 0$ and $hQ(x) = 1$)
- False negatives (FN):
Data points labelled as positive but predicted as negative
($y = 1$ and $hQ(x) = 0$)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive	False Negative
Actual Negative	False Positive	True Negative

Confusion Metrics

- Accuracy is the metric that answers the question “Which fraction of the available items has been correctly classified?”

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision is the metric that answers the question “Which fraction of the items predicted as positive is actually positive?”

$$precision = \frac{TP}{TP + FP}$$

- Recall is the metric that answers the question “Which fraction of the items labelled as positive has been predicted as positive?”

$$recall = \frac{TP}{TP + FN}$$

	Predicted Positive	Predicted Negative
Actual Positive	True Positive	False Negative
Actual Negative	False Positive	True Negative

Overfitting

- A model with the problem of overfitting makes great predictions for training samples but poor ones for validation data. This is because the model adapted itself to every piece of information in the training data until collecting some properties that could be found only within the training data.



Figure 2-16. *Images of cats to train a model*



Figure 2-17. *Test images of cats*

Designing ANN

- Based on the data, draw an expected decision boundary to separate the classes.
- Express the decision boundary as a set of lines. Note that the combination of these lines must yield to the decision boundary.
- The number of selected lines represents the number of hidden neurons in the first hidden layer.
- To connect the lines created by the previous layer, a new hidden layer is added. Note that a new hidden layer is added each time you need to create connections among the lines in the previous hidden layer.
- The number of hidden neurons in each new hidden layer equals the number of connections to be made.

ANN Without Hidden Layer

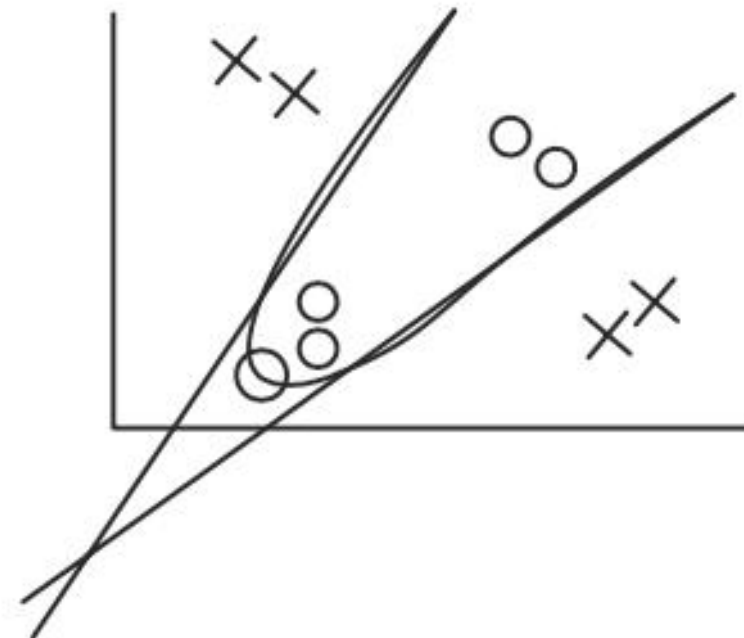
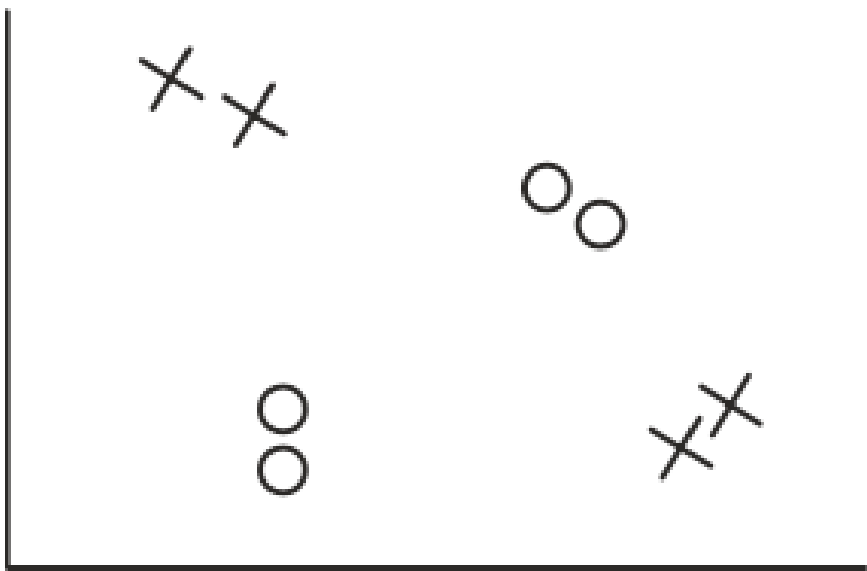


Figure 2-26. Two lines required to classify the problem

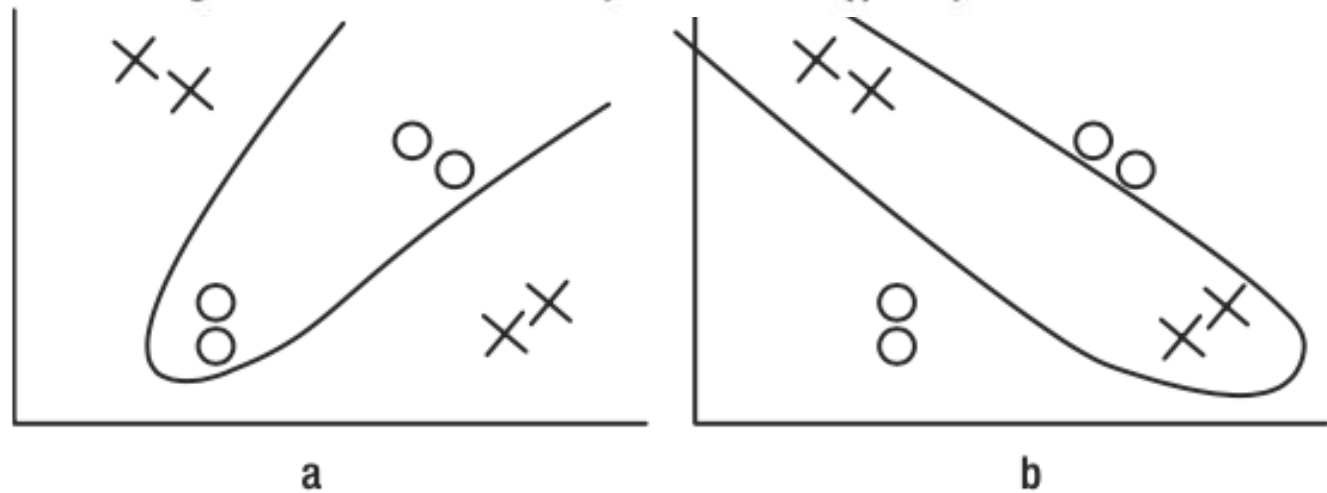


Figure 2-25. Nonlinear classification problem cannot be solved using a single line

ANN With Hidden Layer

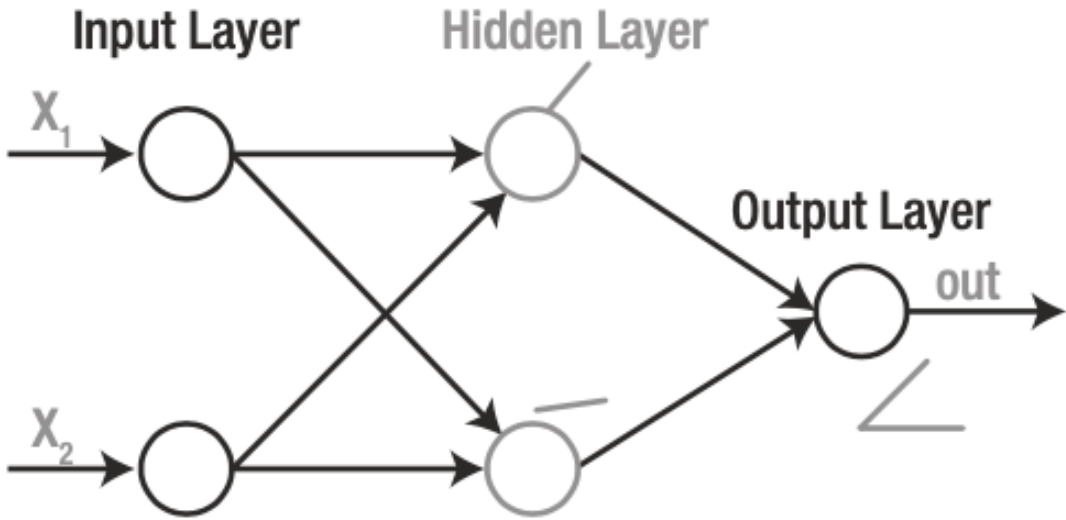
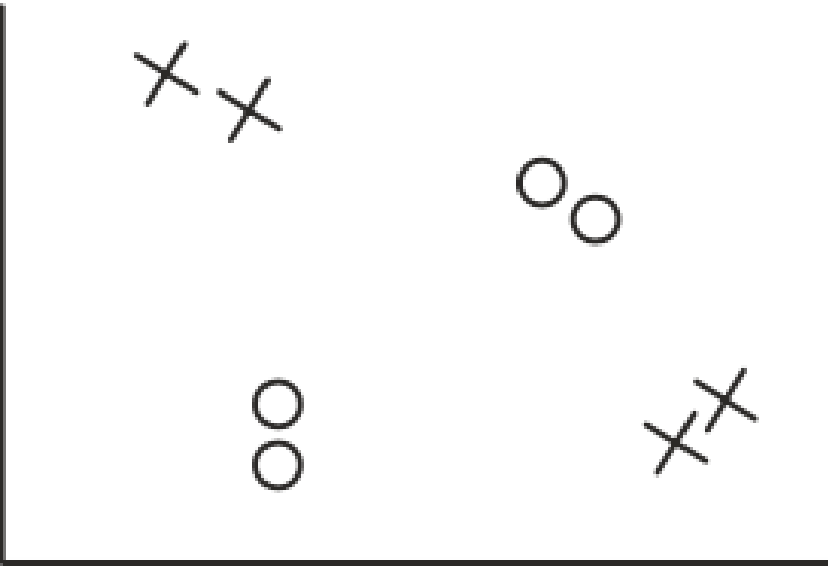


Figure 2-28. Network structure for a classification problem with a curve created by connecting two lines, each one created using a hidden layer neuron

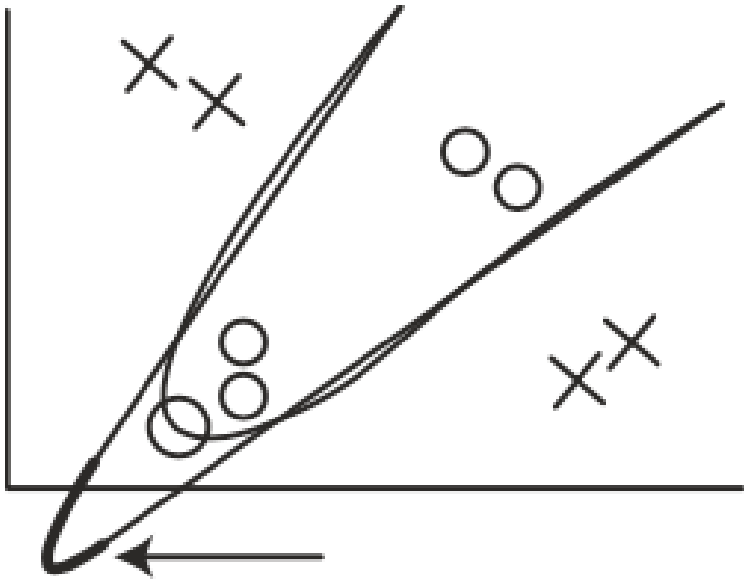


Figure 2-27. Two lines connected to each other using a hidden neuron

Fruits 360 Dataset

- The three pillars for a successful ML application are the data, features, and model.
- Fruits 360 Dataset at <https://www.kaggle.com/moltean/fruits>
- Four classes are selected: Braeburn apple, Meyer lemon, mango, and raspberry.



- Find the most suitable set of features to differentiate these classes.

Fruits 360 Dataset

- Using each pixel as input to the ANN. Each image size is 100×100 pixels. Because the image is color, then there are three existing channels based on the RGB color space: red, green, and blue. Thus, the total number of inputs to the ANN is $100 \times 100 \times 3 = 30,000$.
- The output is 4.
- Assuming there is a single hidden layer of 10,000 neurons, then the total number of parameters in the network is $30,000 \times 10,000 + 10,000 \times 4$, which is more than 300 million parameters.

Convolutional Neural Networks

- ANN is the base of the CNN, with some changes added to make it suitable for analyzing large amounts of data.
- Connecting all neurons together increases the number of parameters even when analyzing very small images.
- Adding more layers and neurons increases the number of parameters. As a result, it is very time-consuming to train such networks even on devices on multiple graphics processing units (GPUs) and multiple central processing units (CPUs).

Convolutional Neural Networks

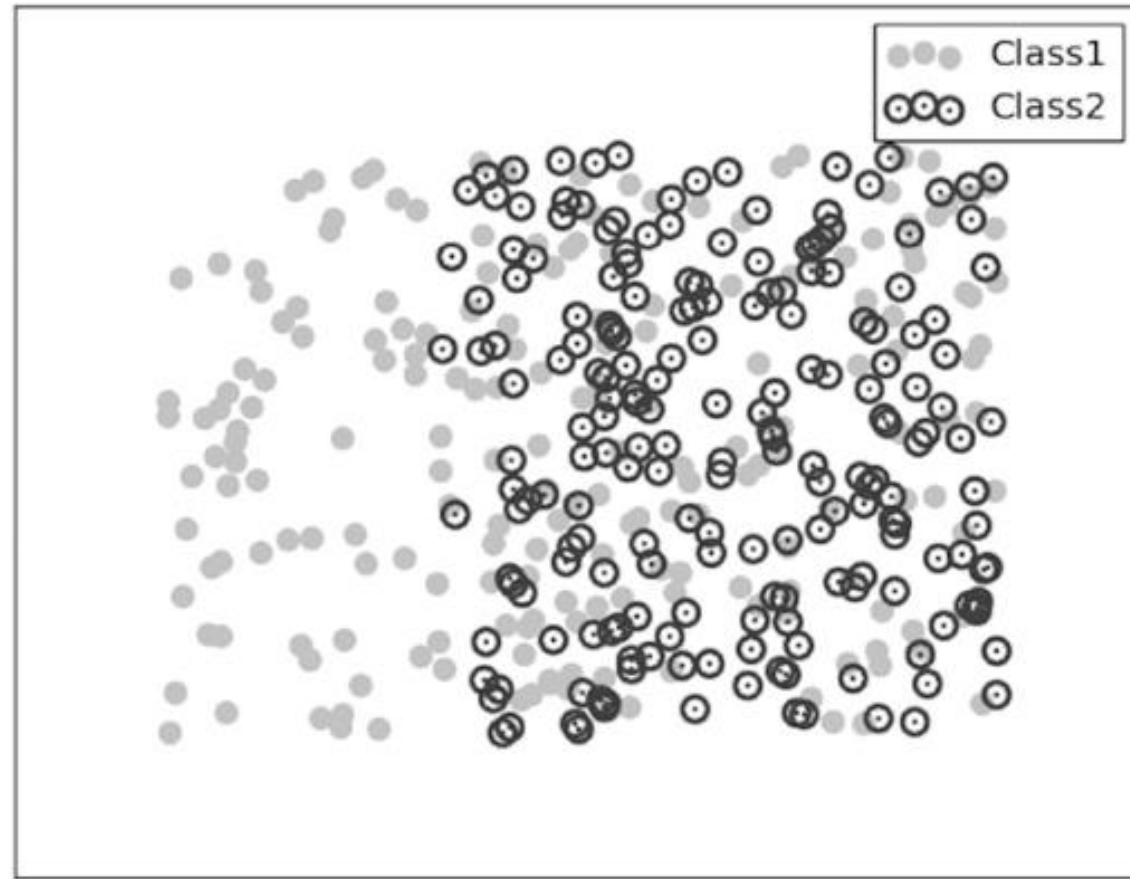


Figure 5-1. Two-class data distribution using $f_1()$. Class 1 samples are represented as filled circles and class 2 samples as empty circles.

Convolutional Neural Networks

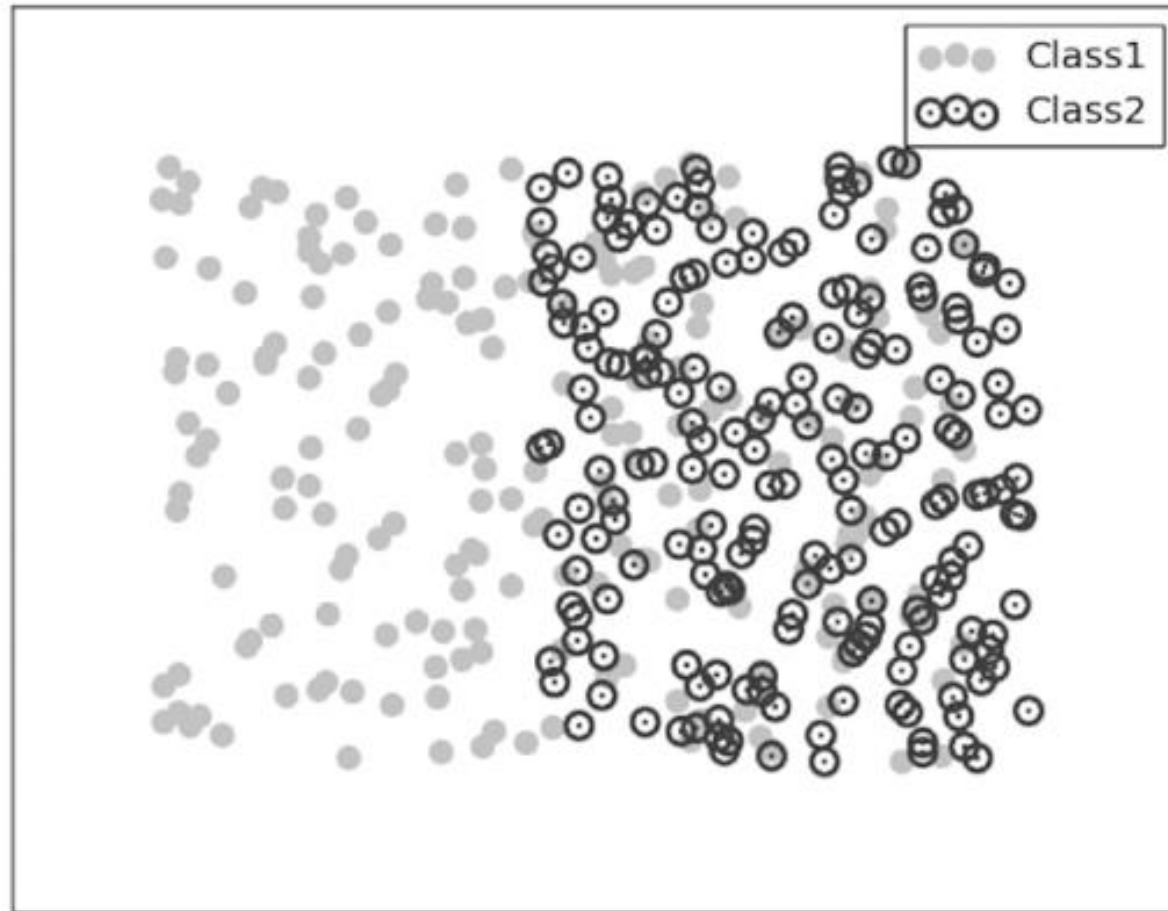


Figure 5-2. Data distribution using $f_2(f_1())$

Convolutional Neural Networks

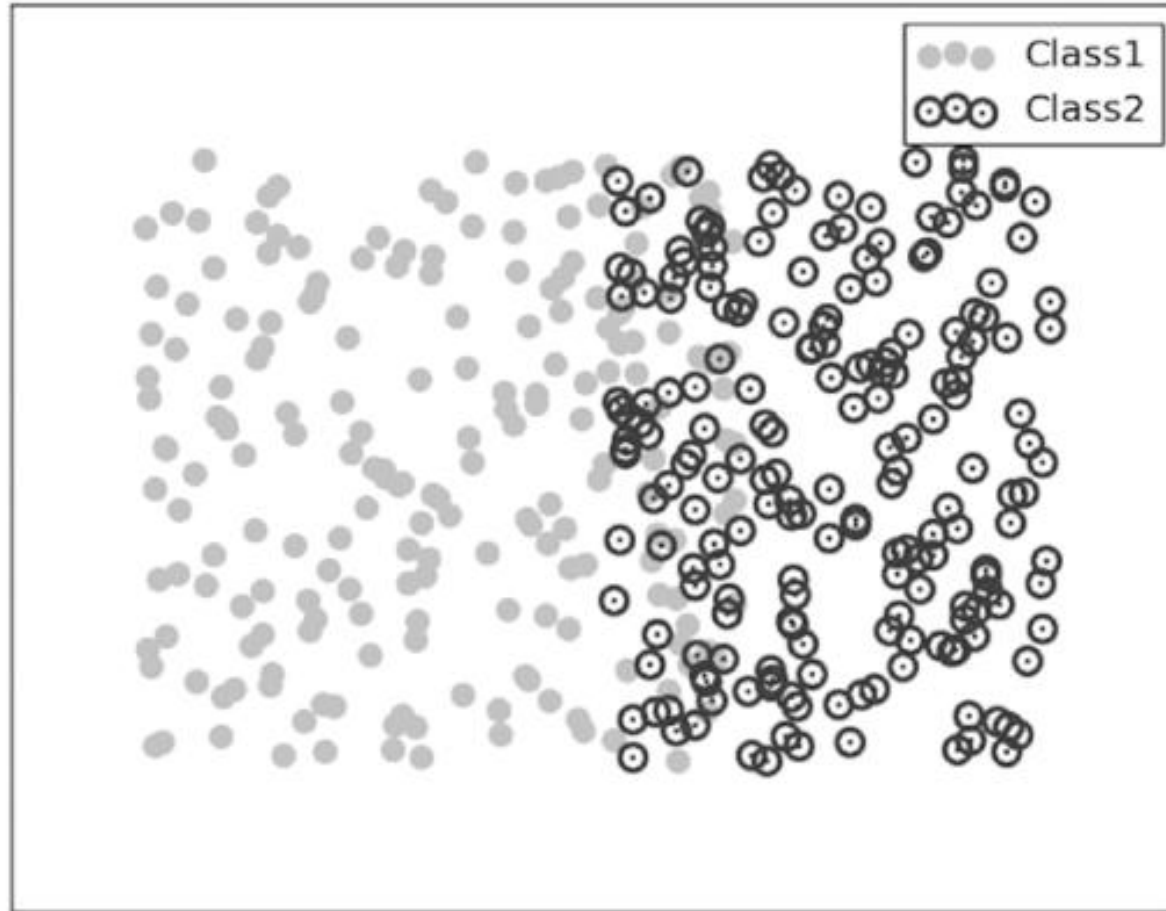


Figure 5-3. Data distribution using $f_3(f_2(f_1()))$

Convolutional Neural Networks

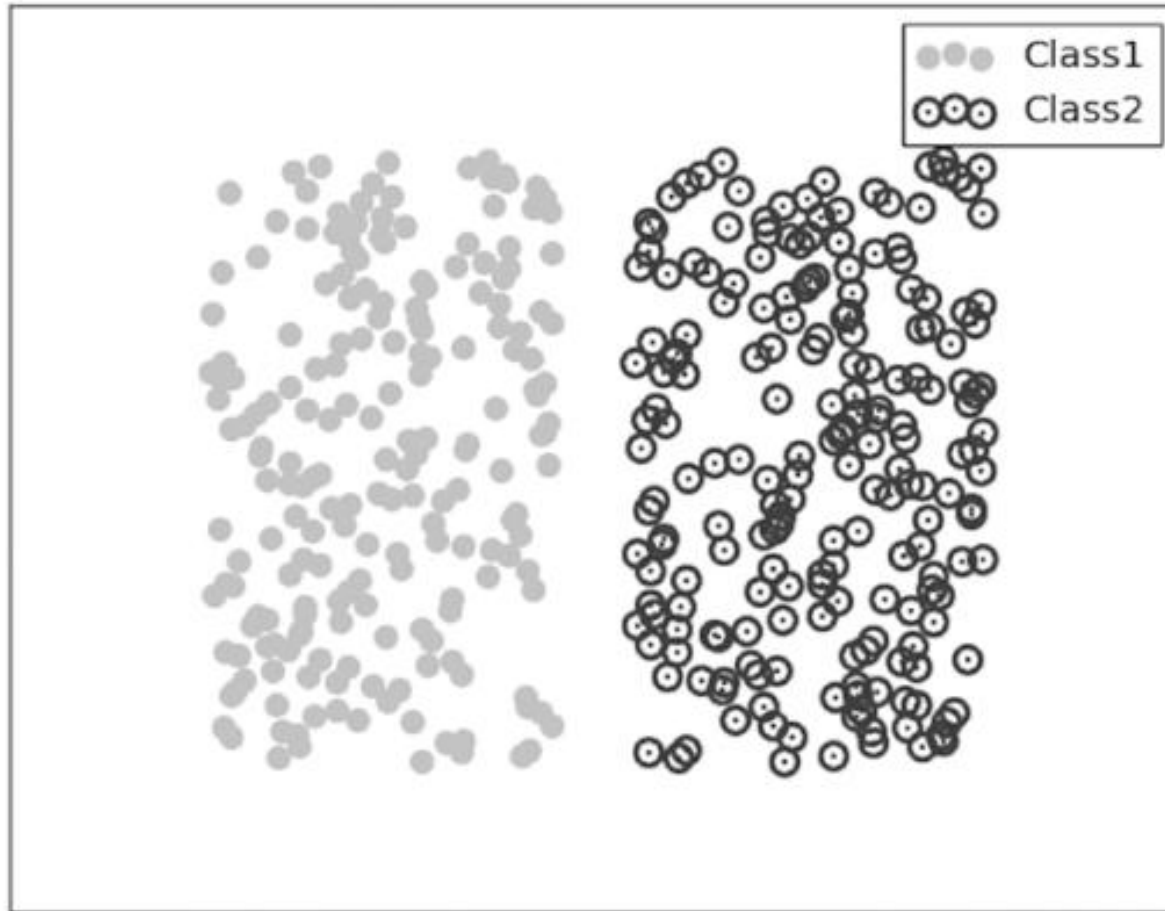


Figure 5-4. *Linear classification after separating the data correctly*

Convolutional Neural Networks

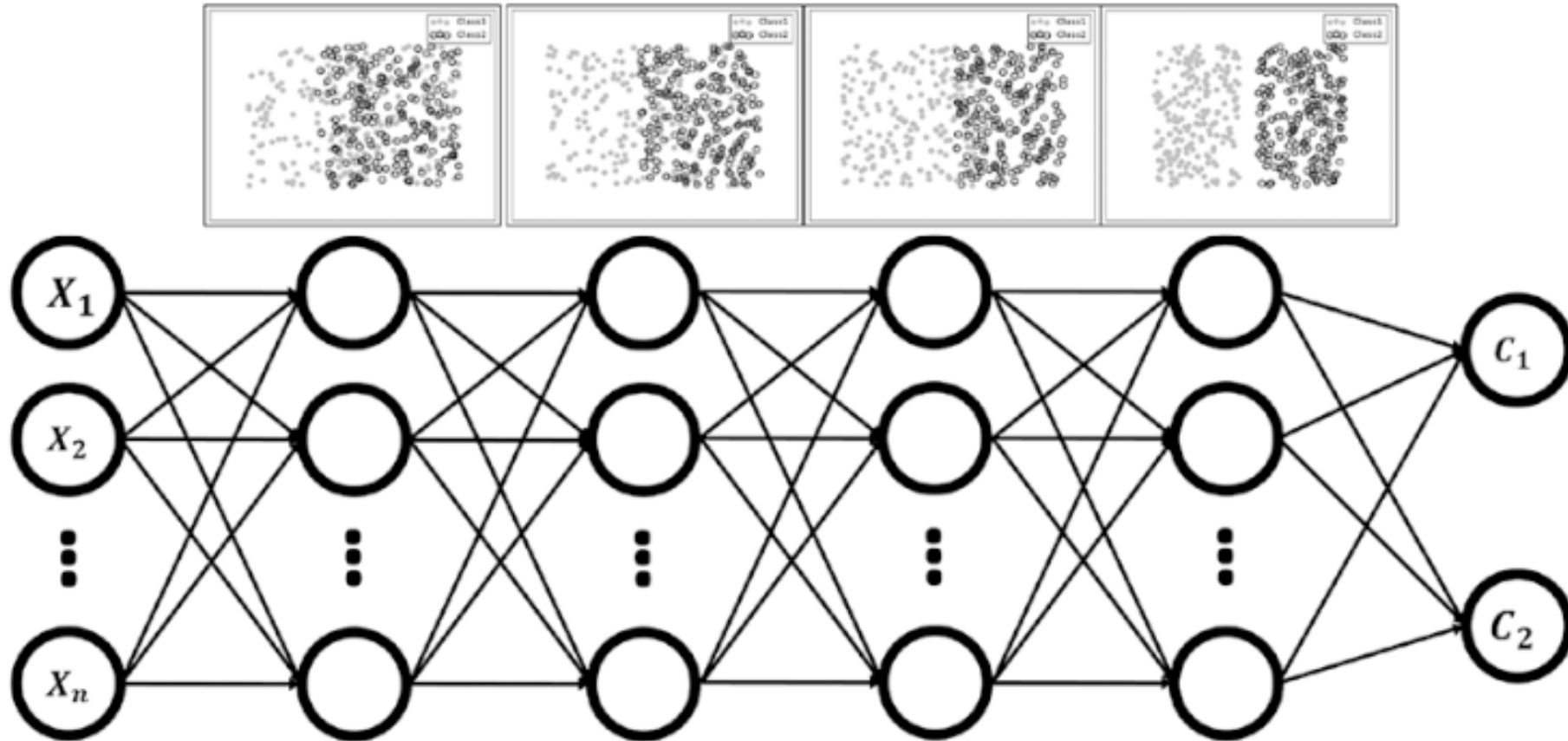
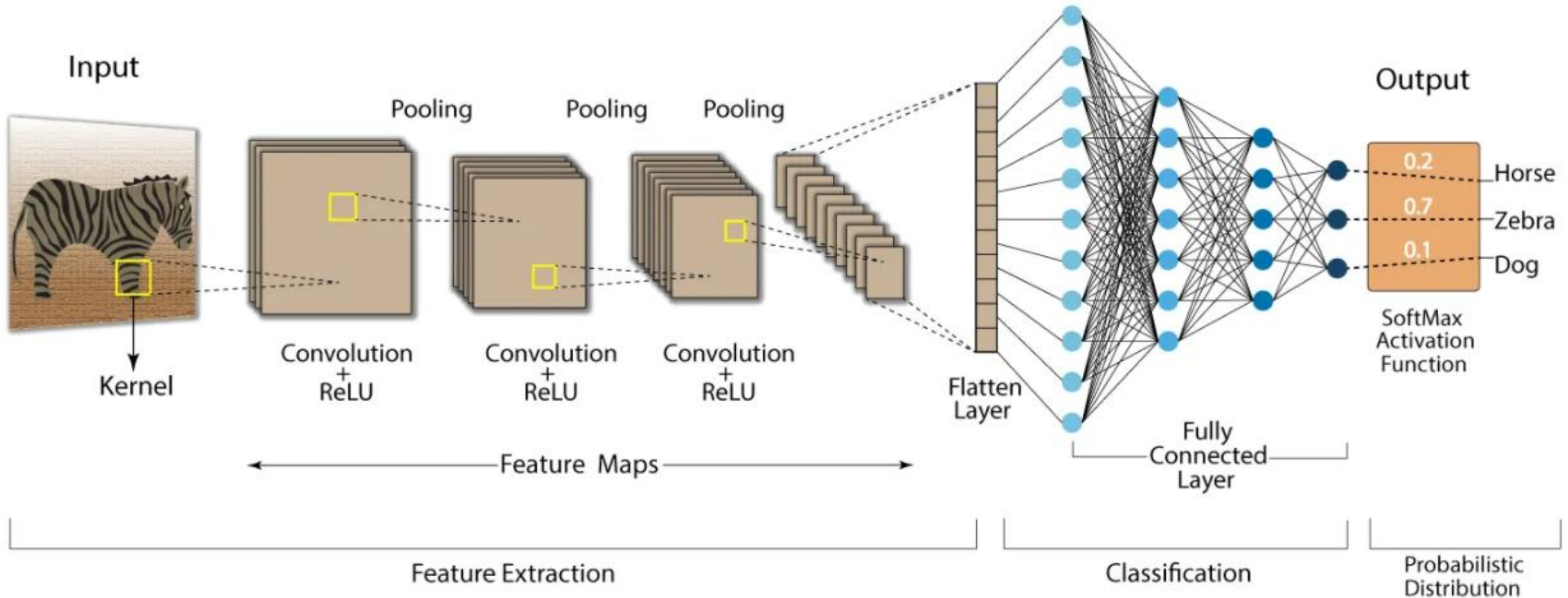


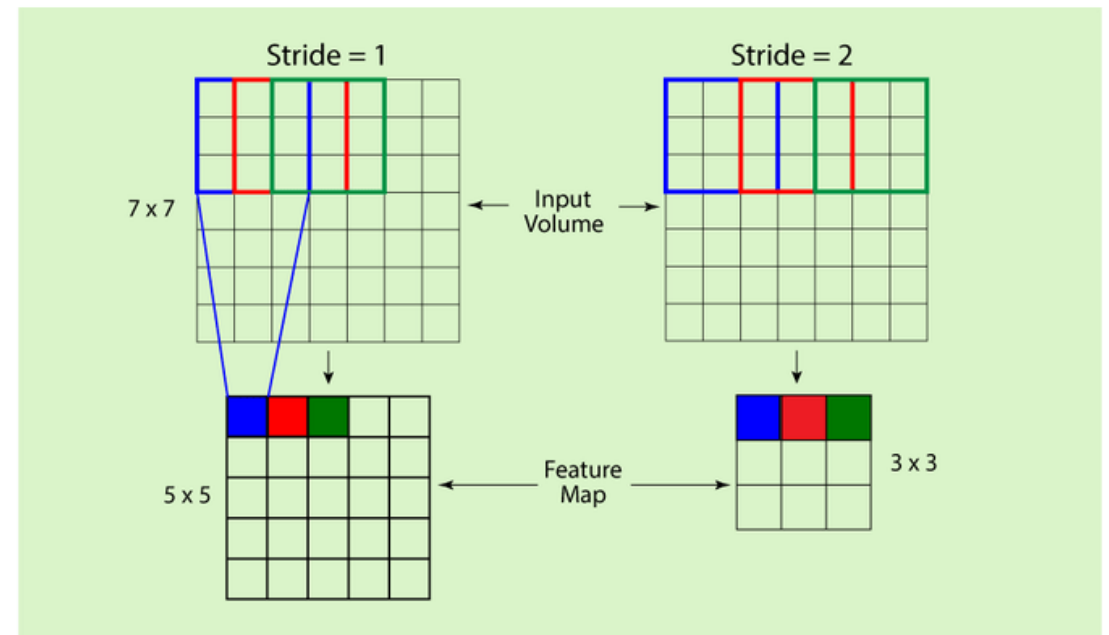
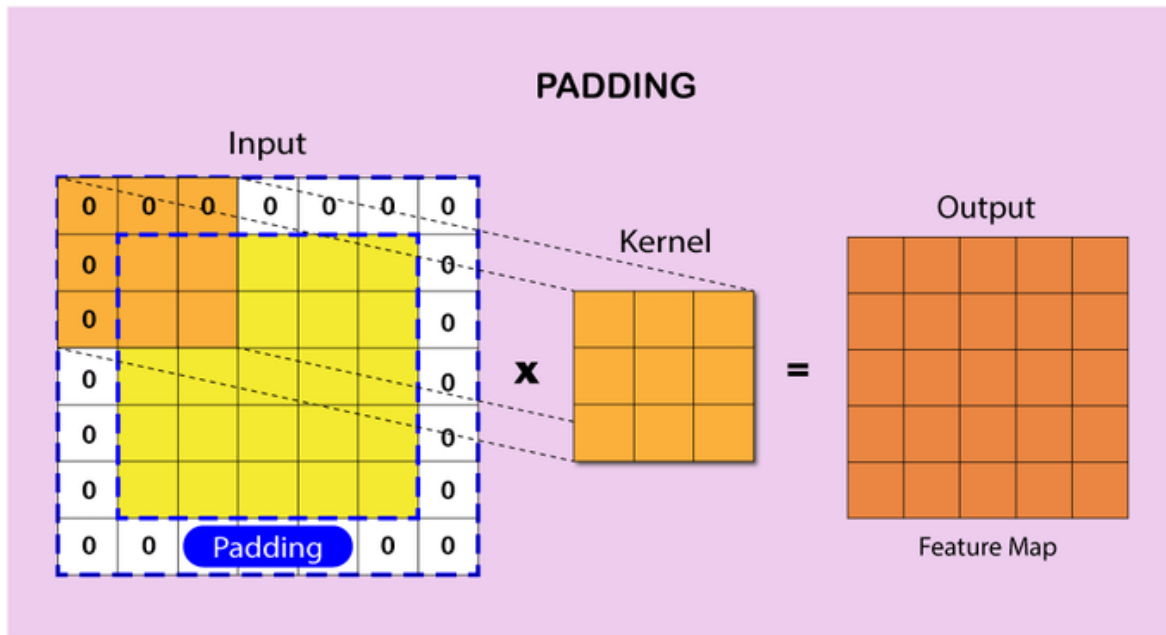
Figure 5-5. ANN required to transform the data for separation of the class by using a linear classifier

Convolutional Neural Networks



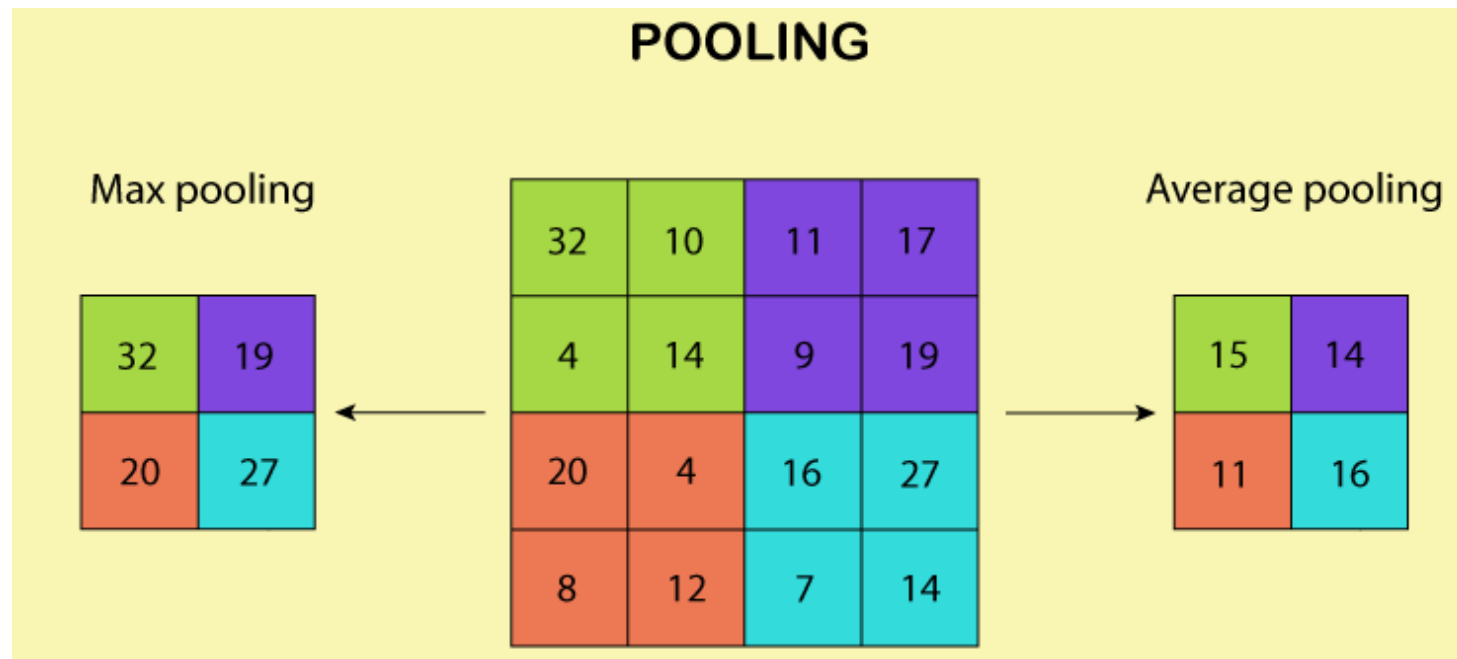
Convolutional Neural Networks

- Convolution Neural Network has input layer, output layer, many hidden layers and millions of parameters that have the ability to learn complex objects and patterns.
- Convolution is the process involving combination of two functions that produces the other function as a result. In CNN's, the input image is subjected to convolution with use of filters that produces a **Feature map**.



Convolutional Neural Networks

- Pooling layer operates on each feature map independently. This reduces resolution of the feature map by reducing height and width of features maps, but retains features of the map required for classification. This is called **Down-sampling**.

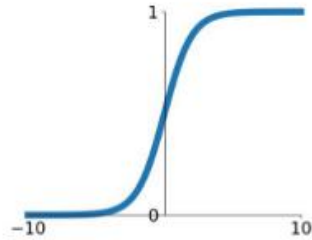


Convolutional Neural Networks

- Activation functions determine the output of a deep learning model, its accuracy, and also the computational efficiency of training a model—which can make or break a large scale neural network.

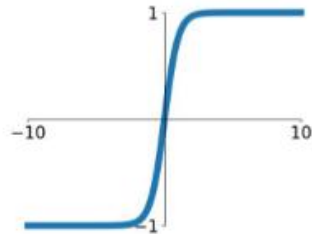
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



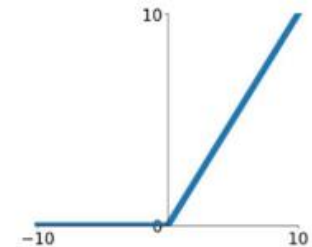
tanh

$$\tanh(x)$$



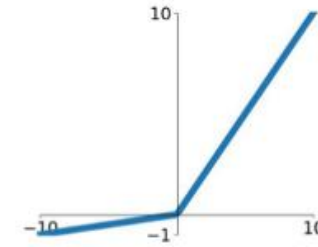
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

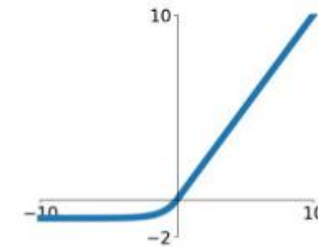


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Read & Discussion

- A Survey of the Recent Architectures of Deep Convolutional Neural Networks
<https://arxiv.org/pdf/1901.06032.pdf>